LLM reads, code verifies

A routing and verification harness for receipt OCR: when a cheap local reader is enough, when to escalate, and how much model error deterministic code catches

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Portfolio & resume

Code & results

1 PROBLEM & SCOPE

I built the receipt-OCR side of a production expense platform (20 employees, 10 corporate cards, about 200 receipts a month) that cut reconciliation time by 90%+, as the sole engineer over 729 merged pull requests.

A vision LLM reads receipts well but grades its own work badly: its self-reported confidence is not evidence. In production, an auto-approval path built on that score was rolled back once review showed what it was.

Question. When is a cheap local reader enough, when should a receipt escalate to a stronger model, and how much of the model's error can deterministic code catch with no ground truth at all?

2 PRODUCTION EVIDENCE

- Built a 90-receipt, 8-currency golden set by hand and benchmarked 7 LLMs (Claude, GPT, Gemini) on accuracy, stability, cost and latency.
- Gemini-to-Claude fallback chain with a fast-fail budget: after a 503 storm pushed one receipt to 87 s, worst case fell to about 10 s.
- +11% amount-extraction pass rate from fixing a validator that silently dropped 12 model-returned fields.
- 40-receipt prompt experiment: the production prompt held 39/40; four rewrites, two-pass extraction and majority voting did not beat it.
- Every model field treated as untrusted input: multilingual injection scan, vendor names accepted only when found in the statement.

3 EXPERIMENT DESIGN

- **300 public receipts** (ICDAR 2019 SROIE, CC-BY-4.0), seed-fixed sample. Exact match on company, date and total after normalization. Scoring is code, never a model.
- **43 synthetic adversarial receipts:** future and ancient dates, arithmetic traps, prompt injection in English and Korean, disguised footers, fullwidth text, non-receipt controls. Printed values and truth are stored separately so a generator slip cannot contaminate a label.
- Gates are tuned on the natural set only; the synthetic set is held out.

4 SYSTEM: A LADDER OF READERS, A WALL OF CHECKS

Tier 1 · local OCR + rules
RapidOCR / Tesseract, CPU. Text and per-token confidence; three fields by rules. $0.

Tier 2 · local VLM
Qwen2.5-VL 3B on Ollama (4 GB GPU). Same prompt and JSON schema as the cloud. $0.

Tier 3 · cloud, cheap
Gemini 3.1 Flash-Lite. $0.25 / $1.50 per 1M tokens.

Tier 4 · cloud, strong
Gemini 3.6 Flash, thinking off. $0.75 / $3.75 per 1M tokens. Vendor failover to Claude.

Code checks on every VLM answer (no ground truth, no model confidence):

- sanitizer: NFKC normalization, EN/KO/JA/ZH injection patterns, field whitelist, type and range
- arithmetic identity: subtotal + tax + tip + discount = total; absent is not zero
- date plausibility: parseable, not future, not ancient
- two-reader agreement: the LLM's total and date must appear in the independent OCR text
- targeted retry: one re-read of the flagged field; code picks the better candidate

Zero flags → **auto-accept**. Flags → escalate one tier → still flagged → **human review**. Model confidence is logged, never used to approve.

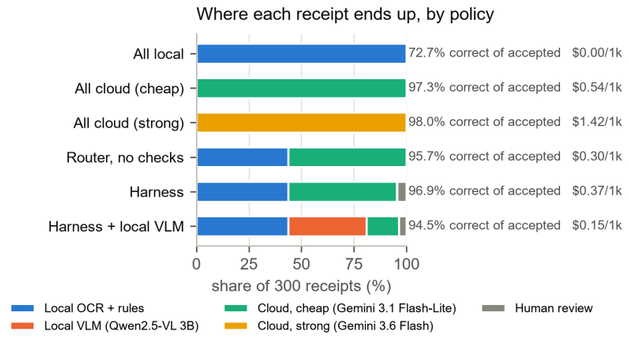


Fig A. Where 300 receipts end up under each policy, with the share of accepted answers whose date and total are correct, and API cost per 1,000 receipts.

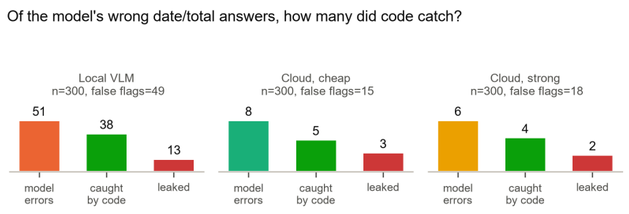


Fig C. Of each model's wrong date/total answers, how many the code checks caught without seeing the answer key.

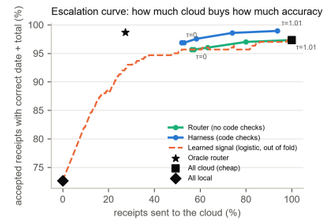


Fig B. Escalation curve: receipts sent to the cloud vs accepted-answer precision as the local confidence threshold τ sweeps. The oracle marks the ceiling a perfect router would reach.

5 RESULTS

READER	DATE + TOTAL	MEDIAN LATENCY	USD / 1K
Local OCR + rules	72.6%	4.7 s	0
Local VLM 3B	82.9%	28.3 s	0
Cloud, cheap	97.3%	1.8 s	0.54
Cloud, strong	98.0%	1.9 s	1.42

ADVERSARIAL SET (CLOUD, CHEAP)	N	MODEL FOOLED	GATE FIRED	LEAKED
Date traps	10	0	10	0
Arithmetic traps	11	1	11	0
Injection, blatant	6	0	6	0
Injection, disguised	10	0	10	0
Controls	6	0	6	0

TARGETED RETRY ON 20 FLAGGED RECEIPTS	FIXED	BROKE	KEPT CORRECT
Blind re-read (no previous values)	0	1	15 / 20
Guided re-read (previous values shown)	1	1	16 / 20

6 FINDINGS

- **Cheap readers get far, not far enough.** On 300 scans, local OCR + rules reads date and total right 73% of the time at $0, a local 3B vision model 83% (28 s on a 4 GB laptop GPU), Flash-Lite 97% at $0.54 per 1,000, 3.6 Flash 98% at $1.42. On 200 phone photos: local OCR 55%, local VLM 81%, cloud 87%.
- **Routing is cheap; the escalation signal is the bottleneck.** An oracle sends 27% of receipts to the cloud for 98.7% at $0.15 per 1,000. The rule router (two readers agree, confidence ≥ 0.9) sends 56% for 95.7%; code checks lift accepted answers to 96.9% at a 4% review load. A logistic model on 13 tier-1 signals (AUC 0.89, out of fold) did not beat the rule on scans and gained 0.5 points on photos: the gap needs richer signals, not a bigger model.
- **Code catches what the model cannot see about itself.** With no answer key, the checks caught 38 of the local model's 51 wrong answers and 5 of Flash-Lite's 8. Of the 18 that slipped through on scans, 16 were a different printed amount (cash tendered, a subtotal), not a misread. On 43 adversarial receipts all three models ignored all 16 injections, every date and arithmetic trap tripped a gate, nothing leaked.

Time budget. Start local, call the cloud at D s: a 30 s budget on this laptop sends 43% to the cloud for 96.3% at $0.23 per 1,000; a 4x faster GPU with a 10 s budget sends 19% for 94.4%. **Next.** 10k receipts across SROIE, CORD and WildReceipt; richer escalation signals (layout, a cheap local VLM pass); per-currency tolerances; drift monitoring under a spend cap.

Sources. ICDAR 2019 SROIE (Huang et al., arXiv:2103.10213), jsdnrs mirror on Hugging Face, CC-BY-4.0 · Tesseract 5.4, Apache 2.0 · RapidOCR / PP-OCR, Apache 2.0 · Qwen2.5-VL, Apache 2.0, served by Ollama · Google Gemini API pricing page, accessed 2026-10-31 · Fallback chain and sanitizer adapted from the author's open-source boda project (MIT). Production figures are from the author's 2026 engagement; the platform itself is private.

Experiment run 2026-10-31 to 11-01 · code, data manifest and all results are in the repository